

"Techniques like **focused ultrasound or radiofrequency ablation are non-invasive** and can be used to reduce liver fibrosis"

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changes—increased physical activity, improved diet and, in some cases, weight-loss surgery. Weight loss, in particular, has been shown to have a powerful effect on reducing liver fat. Typically, losing 10 per cent of your body weight or more is recommended. In a 2015 study in the *Journal of Clinical and Experimental Hepatology*, central obesity was found to be the most common risk factor in 84 per cent of the Indian patients with NAFLD. Doctors, therefore, recommend not just overall but abdominal weight loss for Indians to reverse NAFLD.

The food you eat plays a huge role in treatment. For good liver health, it is important to include Omega-3 fatty acids, antioxidants, Vitamin E and enough fibre in your diet, says Twincy Ann Sunil, clinical dietitian at Apollo Spectra, Koramangala, Bengaluru. Good sources of Omega-3 fatty acids are walnuts, flax seeds, salmon and mackerel fish; nuts and oilseeds for Vitamin E; and (carefully chosen) fruits and vegetables for fibre. "Use cold-pressed oil for cooking, as some refined oil might have higher amount of transfat due to the processing methods used. Instead of snacking on chips and bakery items, snack on veggie salads or a chia seed pudding made with yoghurt and topped with fruits, salted nuts," she adds.

Dr Yogesh Chawla, former director and head, department of hepatology, Postgraduate Institute of Medical Education and Research, Chandigarh, suggests running a media campaign every 15 days for public awareness of causes of fatty liver. The campaign, he emphasises, should focus on key dietary habits: "Eat only when hungry; snacking is not a solution for loneliness or insecurity; choose whole foods rather than processed foods; eat according to yogic wisdom—stop when 80 per cent full; start your meal with a glass of water followed by a bowl of raw salads; drink water first whenever you have a feeling of hunger. False hunger disappears. Real hunger stays."

However, when fatty liver progresses to an advanced stage or presents with other complications, medications or procedures may be recommended. "Medications are given for management of diabetes or cholesterol, diuretics to manage fluid build-up in the body, beta blockers to reduce hypertension in the portal vein (that leads to the liver) and antibiotics if there is liver disease," says Dr Somnath Chattopadhyay, consultant and head of department, hepato-pancreato-biliary surgery and liver transplant, Kokilaben Dhirubhai Ambani Hospital, Mumbai. If needed, procedures at this stage can include draining excess fluid from the abdomen, and stopping bleeding from enlarged blood vessels in the oesophagus.

While liver cirrhosis, the final stage of liver damage, is irreversible, newer treatment options can reverse the stage before it: liver fibrosis, or when scar tissues form on the liver. "Some promising areas include anti-fibrotic drugs that target specific molecules involved in the scarring process. Although many drugs are still under development, some like Saroglitazar are available in India," says Dr Chattopadhyay. Cell therapy to regenerate damaged liver tissue, gene therapy to modify genetic liver defects or non-invasive therapies are also becoming common. "Techniques like focused ultrasound or radiofrequency ablation are non-invasive and can be used to reduce liver fibrosis," adds Dr Chattopadhyay.

However, once the liver is damaged, patients have no option but to go for a transplant. Currently, the one-year survival rate of a liver transplant is above 90 per cent in many hospitals, compared to 50 per cent in the 1980s. Technology and advanced surgical techniques have played a very important role in this. "We have improved surgical techniques, including minimally invasive approaches, better anaesthesia equipment, which reduce complications and mortality. Additionally, the development of more effective and less toxic immunosuppressive medications has helped prevent organ rejection. Advances in organ preservation and allocation have led to the use of higher-quality donor organs," says Dr Chattopadhyay. According to various estimates, the annual demand for liver transplants is between 25,000 and 30,000 in India. In 2022, some 2,900 living donor liver transplants and 760 deceased donor liver transplants were performed in India. In that same year, India became the third largest liver transplant-performing country in the world after the US and China. Doctors, however, say there is still a very large unmet need.

And while improved organ donation and transplant surgeries for the liver may be good news, the fact that the need for them is rising is not a good sign, say doctors. In India, nearly 66 per cent of all deaths are on account of non-communicable diseases, and fatty liver is both a high-risk factor for them as well as their victim. The real good news will be a reduction in the figures for the disease, particularly among young children, for whom it can spell a lifetime sentence of poor health. Public awareness of the signs and risks of liver disease can help diagnose the condition in time, but only a complete overhaul of lifestyle, where we take ownership of and prioritise our health through exercise, a healthy balanced diet and regular preventive screenings can bring about significant change. ■